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What is an anomaly?

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Question: Can we characterize anomaly just as low-probability events?

Answer: No. The probability of any event, described with sufficient precision within its context of occurrence, is zero! Maybe anomaly has to do with complexity...

Anomaly Detection



alternative.

One could say that finding such

simple features automatically is a

difficult task.

Indeed it is!

This is a big challenge for artificial

intelligence.

But conversely,

we wouldn't consider a system as

intelligent if it is blind to features

revealing anomalies that are obvious to us.

Anomaly Detection is another typical problem in Unsupervised Learning. Anomalies of interest in Machine Learning are **outliers**. The problem of detecting outliers is close to the clustering task, in the sense that the learner tries to infer a model that describes "normal" data. Contrary to clustering, however, the point of anomaly detection is not to group similar data but to discard data that do not fit the model.

We use the same previously-mentioned **K-Means prototype-based model**. The complexity criterion, however, will be used for each data point to decide whether it is well described by the model. The compression rate is defined as $\frac{\text{complexity after}}{\text{complexity before}}$. If it is close to 1, the point does not fit with the model and may be regarded as an anomaly.

Anomaly detection 1

1/1 point (ungraded)

Modify the `anomalyDetection` function in order to select abnormal points. A point is considered abnormal if the compression rate exceeds the threshold value. The function should return a list of indices. Use the numpy method `where`. Copy the 'return' line here.

```
return np.where(compressionRates > threshold)
```



Answer: `return np.where(compressionRates > threshold)`

```
np.where(compressionRates > threshold)
```

Anomaly detection 2

0/1 point (ungraded)

Choose the dataset 'diabetes', and set the number of prototypes to the best value previously found. Uncomment the last lines to call `anomalyDetection`. Determine the threshold value that will select the 9 most abnormal points.

Which value did you find?

0.83

✗ Answer: 0.9

0.83

Explication

With a threshold value of 0.9 and K=42, we get 9 exceptions.

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Anomaly detection 3

1/1 point (ungraded)

The preceding exercises aimed at detecting outliers. What do you think...?

☒ An outlier is an anomaly because it is too simple.

☐ An outlier is an anomaly because it is too complex.



Answer

Correct:

In our prototype-based clustering, outliers couldn't be compressed because they were too far away from any cluster center. So they remain complex. However, being complex offers a simple characterization. So if we select 10 abnormal data point, the complexity of these points is not much more than 4 bits (see the "ugly duckling paradox" in Chapter 1.)

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